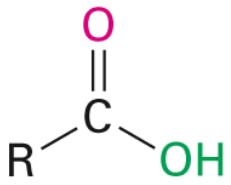
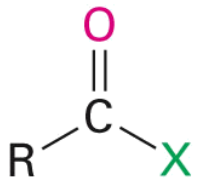
Chapter 21
Carboxylic Acid Derivatives: Nucleophilic Acyl Substitution Reactions

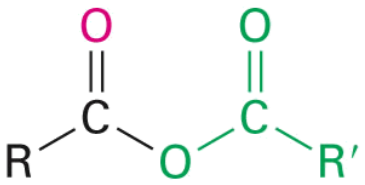
Carboxylic Compounds



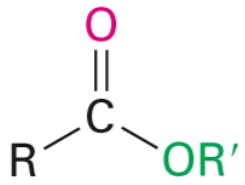
Carboxylic acid



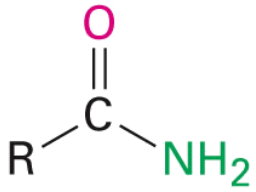
Acid halide
(X = Cl, Br)



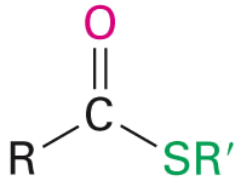
Acid anhydride



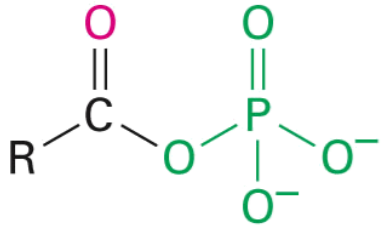
Ester



Amide



Thioester

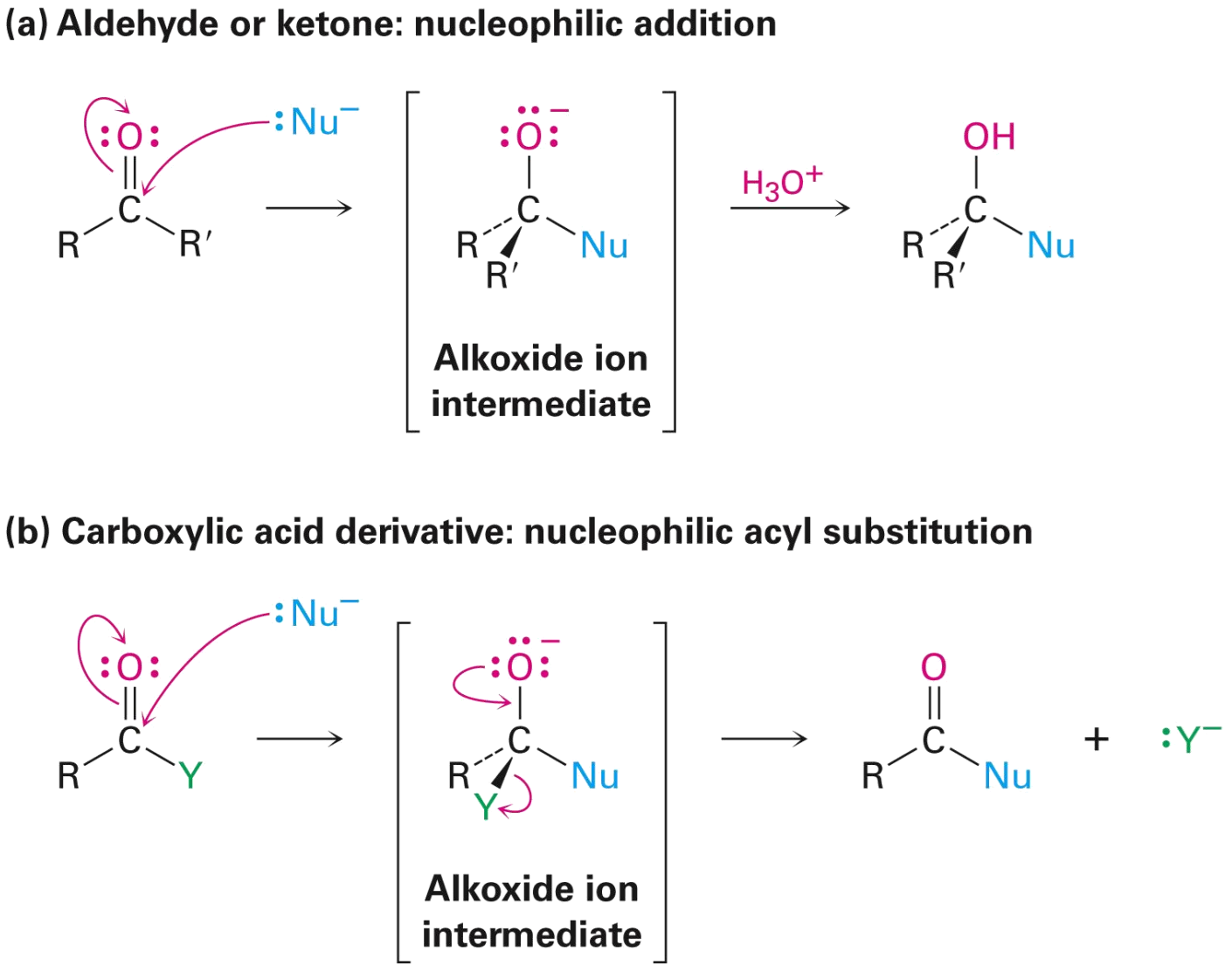


Acyl phosphate

Nucleophilic Acyl Substitution Reactions

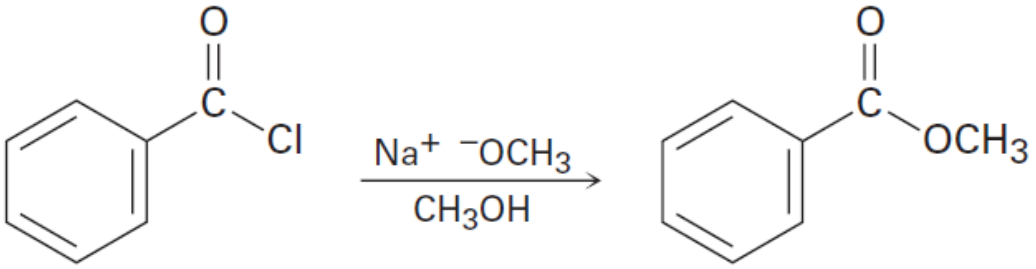
- Tetrahedral intermediate eliminates one of the two substituents originally bonded to the carbonyl carbon
  - Leads to a net **nucleophilic acyl substitution reaction**
- Carboxylic acid derivatives have an acyl carbon bonded to a group –Y that can leave

Figure 21.1 - The General Mechanisms of Nucleophilic Addition and Nucleophilic Acyl Substitution Reactions

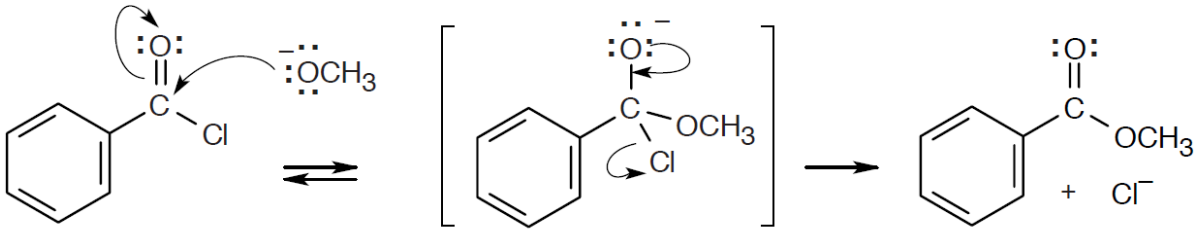


Worked Example

- Show the mechanism of the following nucleophilic acyl substitution reaction
  - Use curved arrows to indicate the electron flow in each step



- Solution:

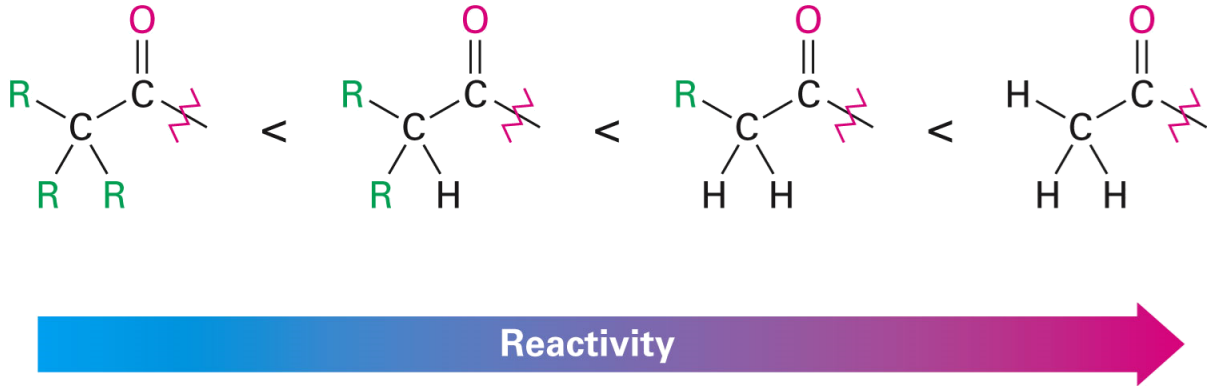


addition of methoxide to form a tetrahedral intermediate

elimination of Cl^-

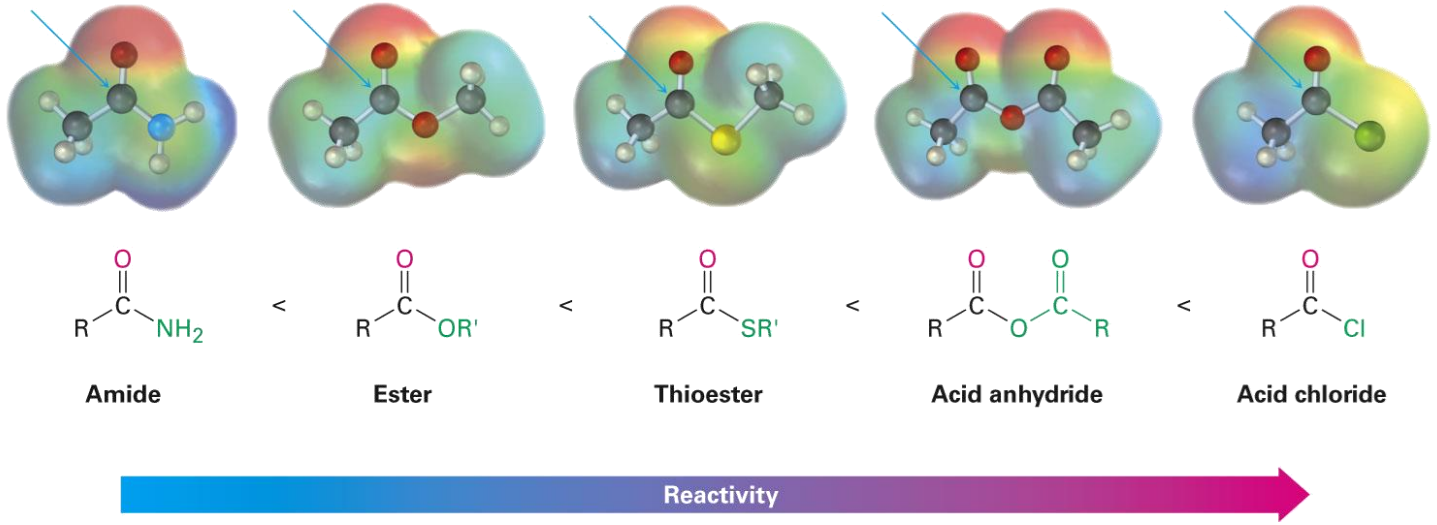
Nucleophilic Acyl Substitution Reactions

- Relative reactivity of carboxylic acid derivatives
  - Nucleophiles react more readily with unhindered carbonyl groups



  - Electrophilic carbonyl groups are more reactive to addition
  - The intermediate with the best leaving group decomposes fastest

Nucleophilic Acyl Substitution Reactions



- Various substituents affect the polarization of a carbonyl group
  - Similar to the way they effect reactivity of an aromatic ring toward electrophilic substitution

Nucleophilic Acyl Substitution Reactions

- A more reactive acid derivative can be converted into a less reactive one
- Acid halides and acid anhydrides react rapidly with water

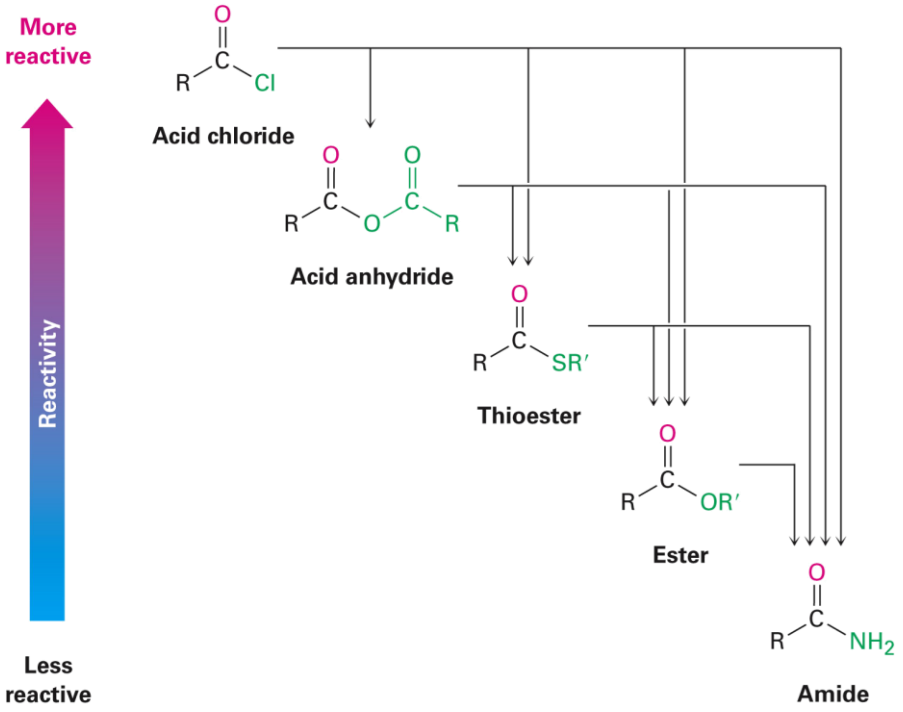
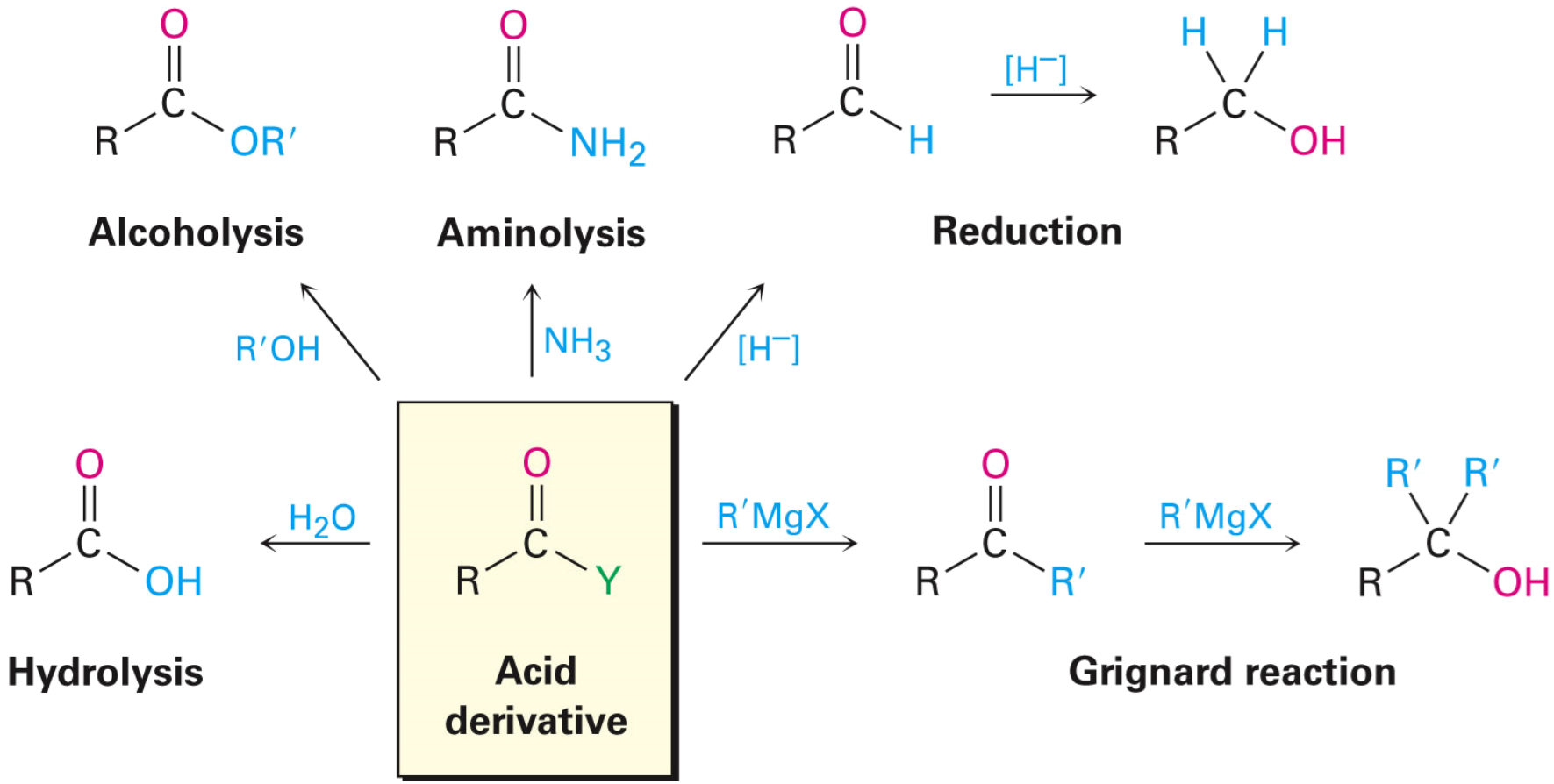
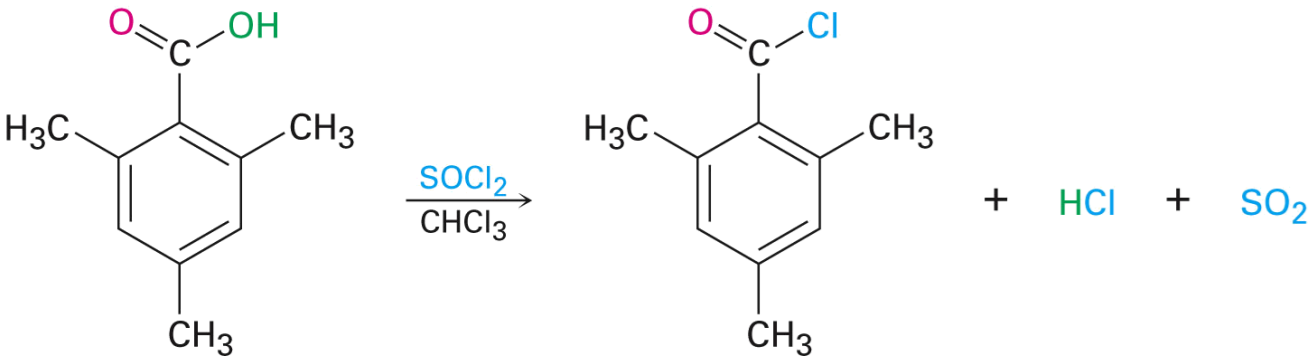


Figure 21.3 - Some General Reactions of Carboxylic Acid Derivatives



Reactions of Carboxylic Acids

- Conversion of carboxylic acids into acid chlorides
  - Reaction with thionyl chloride, $SOCl_2$

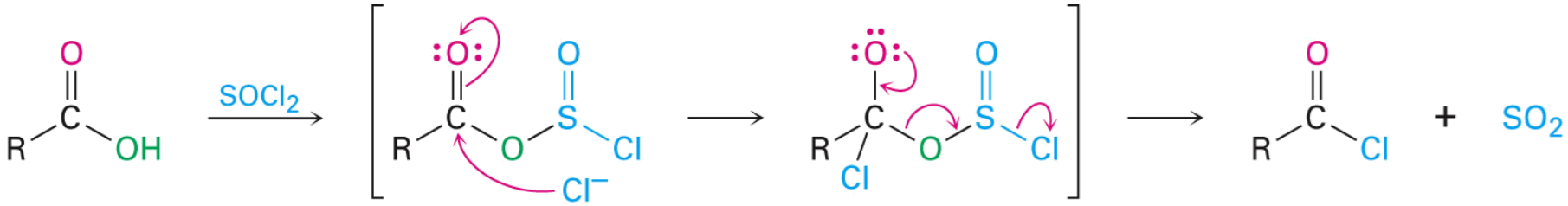


2,4,6-Trimethylbenzoic acid **2,4,6-Trimethylbenzoyl chloride (90%)**

  - Carboxylic acid is first converted into an acyl chlorosulfite intermediate which replaces the –OH of the acid with a much better leaving group
  - Chlorosulfite then reacts with a nucleophilic chloride ion

Reactions of Carboxylic Acids

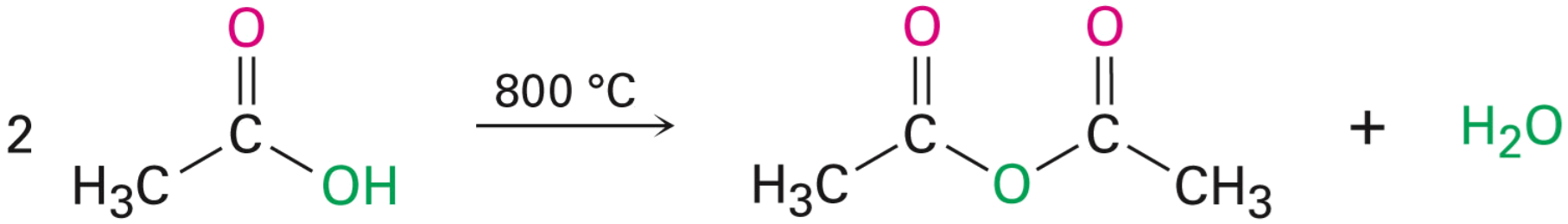
- Occurs by a nucleophilic acyl substitution pathway
- Carboxylic acid is converted into a chlorosulfite which then reacts with chloride



Carboxylic acid **A chlorosulfite** **Acid chloride**

Reactions of Carboxylic Acids

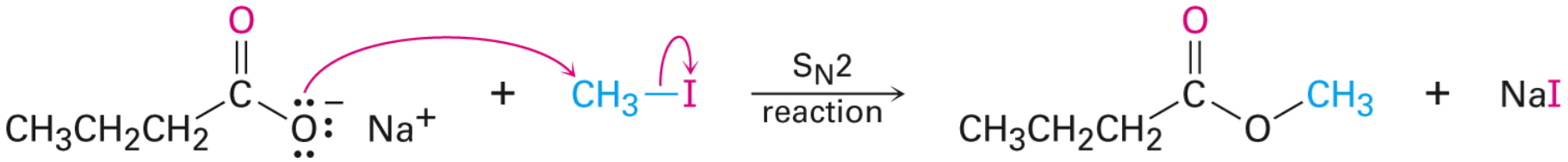
- Conversion of carboxylic acids into acid anhydrides
  - Acid anhydrides can be derived from two molecules of carboxylic acid by heating to remove water



Acetic acid **Acetic anhydride**

Reactions of Carboxylic Acids

- Conversion of carboxylic acids into esters
  - Through reaction of a carboxylate anion with a primary alkyl halide

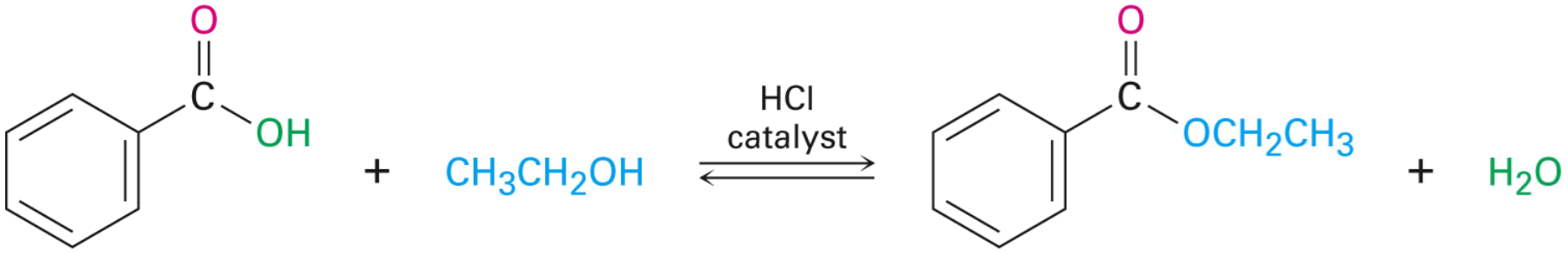


Sodium butanoate

Methyl butanoate (97%)

Reactions of Carboxylic Acids

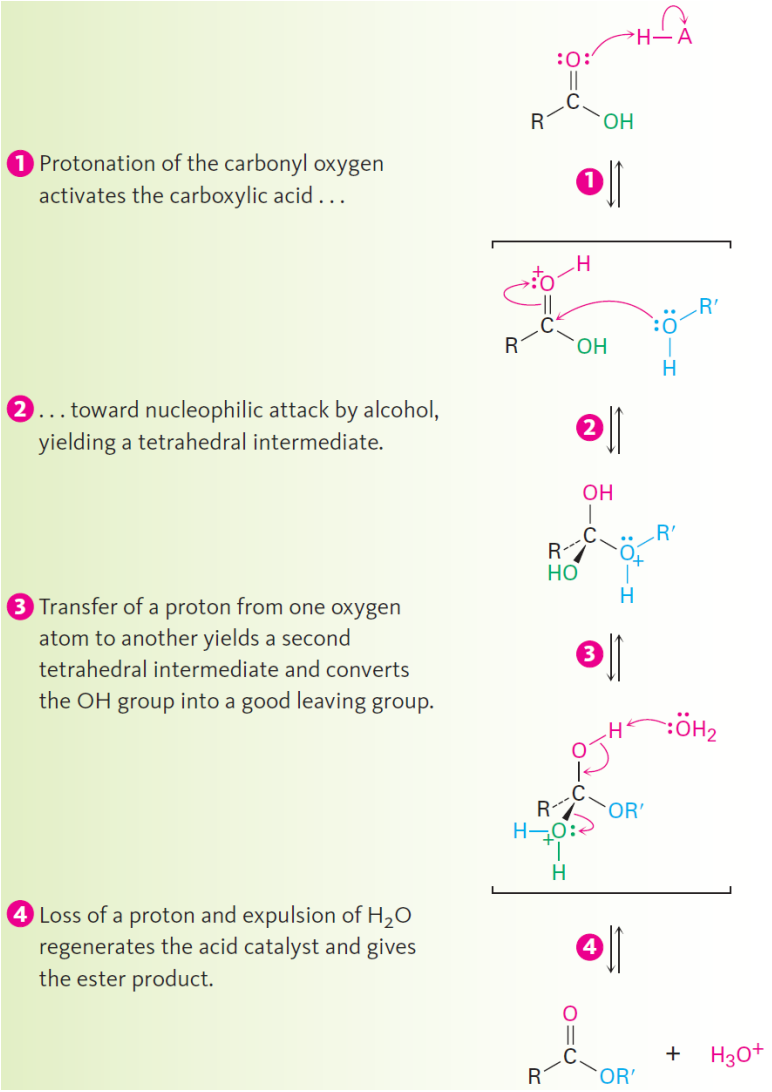
- **Fischer esterification reaction**: Synthesis of esters by an acid-catalyzed nucleophilic acyl substitution reaction of a carboxylic acid with an alcohol



Benzoic acid **Ethyl benzoate (91%)**

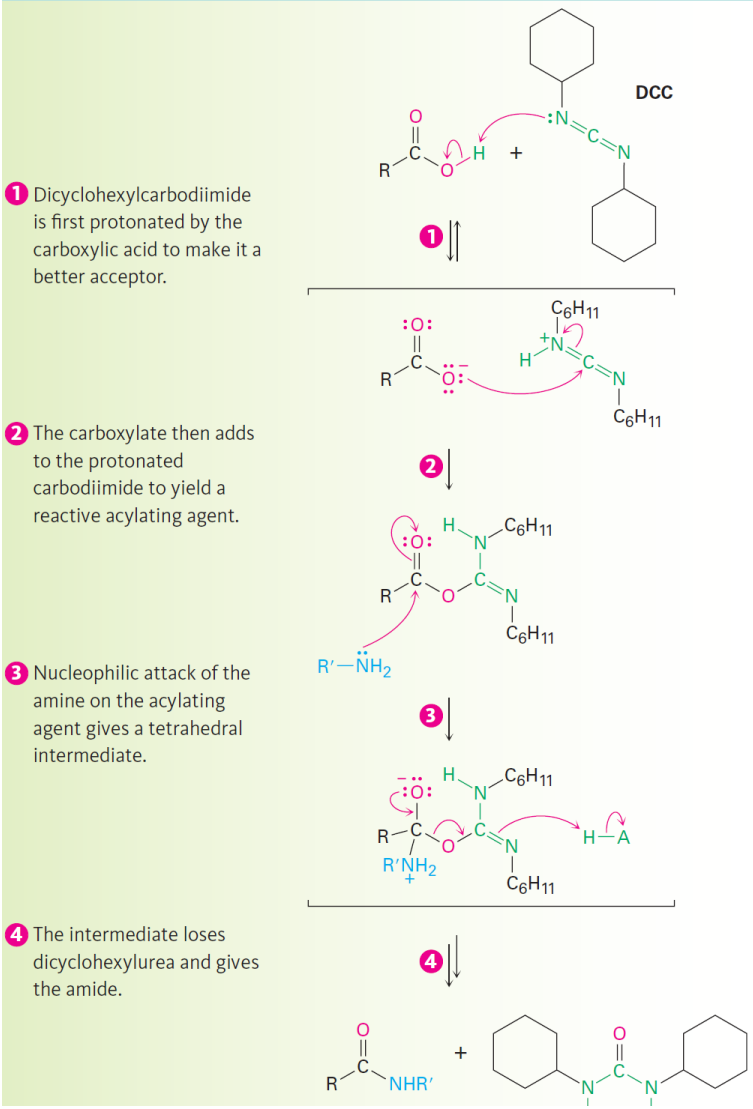
Mechanism of the Fischer Esterification

- ^{18}O-labeled methanol reacts with benzoic acid, the methyl benzoate produced is ^{18}O-labeled but the water produced is unlabeled



Reactions of Carboxylic Acids

- Conversion of carboxylic acids into amides
  - Difficult to prepare by direct reaction of carboxylic acids
  - Prepared by activating the carboxylic acid with dicyclohexylcarbodiimde, followed by addition of the amine



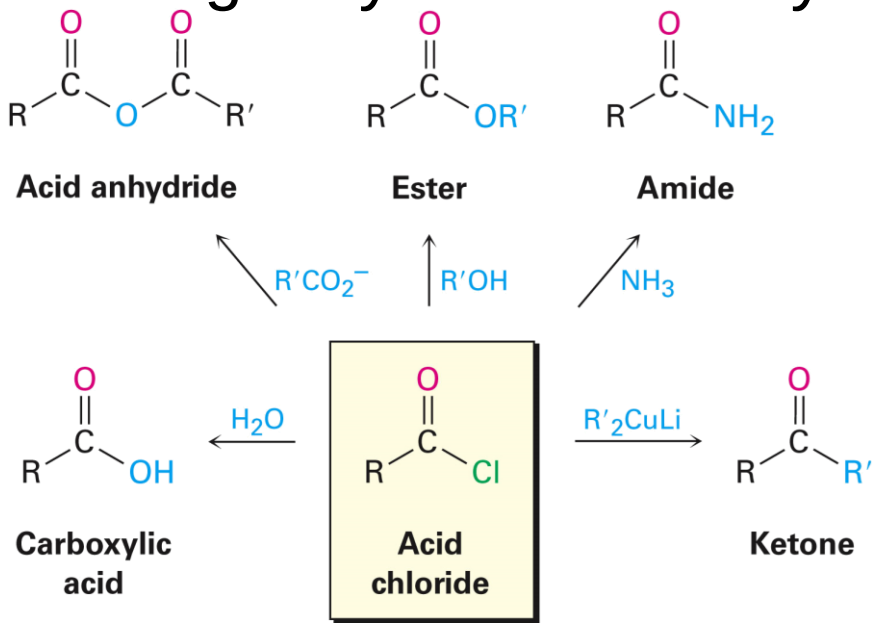
Chemistry of Acid Halides

- Preparation of acid halides
  - Acid chlorides are prepared from carboxylic acids by reaction with $SOCl_2$
  - Reaction of a carboxylic acid with PBr_3 yields the acid bromide



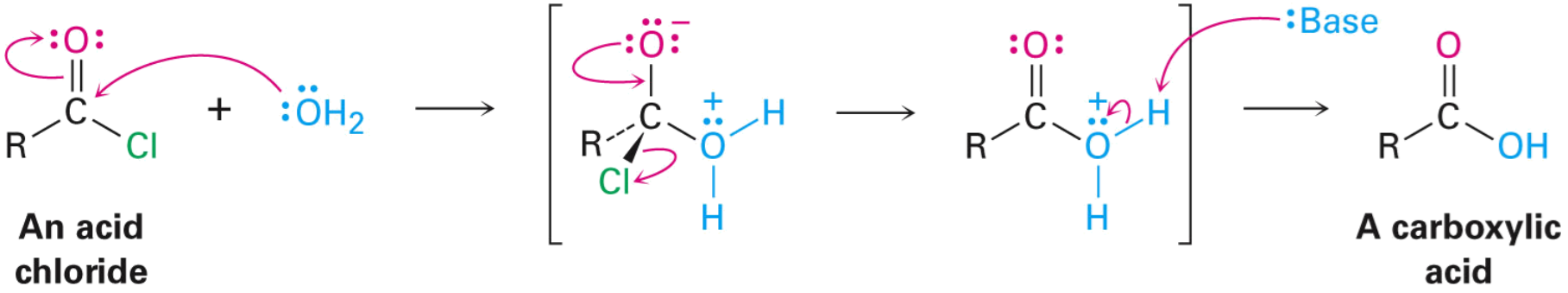
Chemistry of Acid Halides

- Reaction of acid halides
  - Nucleophilic acyl substitution mechanisms
  - Halogen replaced by –OH, by –OR, or by –NH_2
  - Reduction yields a primary alcohol
  - Grignard reagent yields a tertiary alcohol



Chemistry of Acid Halides

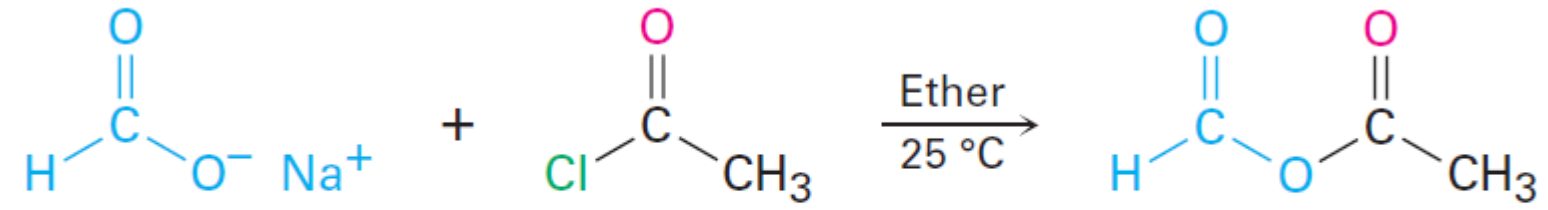
- Conversion of acid halides into acids: Hydrolysis
  - Acid chlorides react with water to yield carboxylic acids



  - HCl is generated during the hydrolysis: A base is added to remove the HCl

Chemistry of Acid Halides

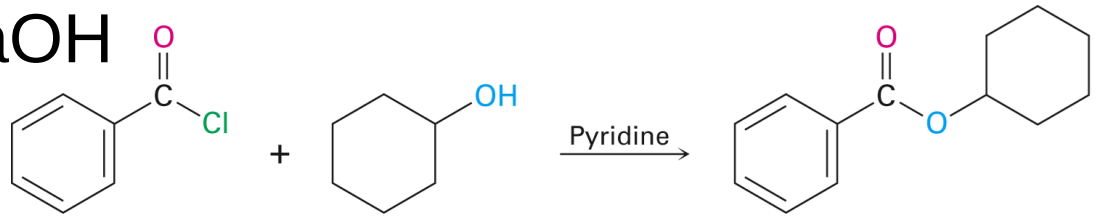
- Conversion of acid halides into anhydrides
  - Nucleophilic acyl substitution reaction of acid chloride with a carboxylate anion gives acid anhydride



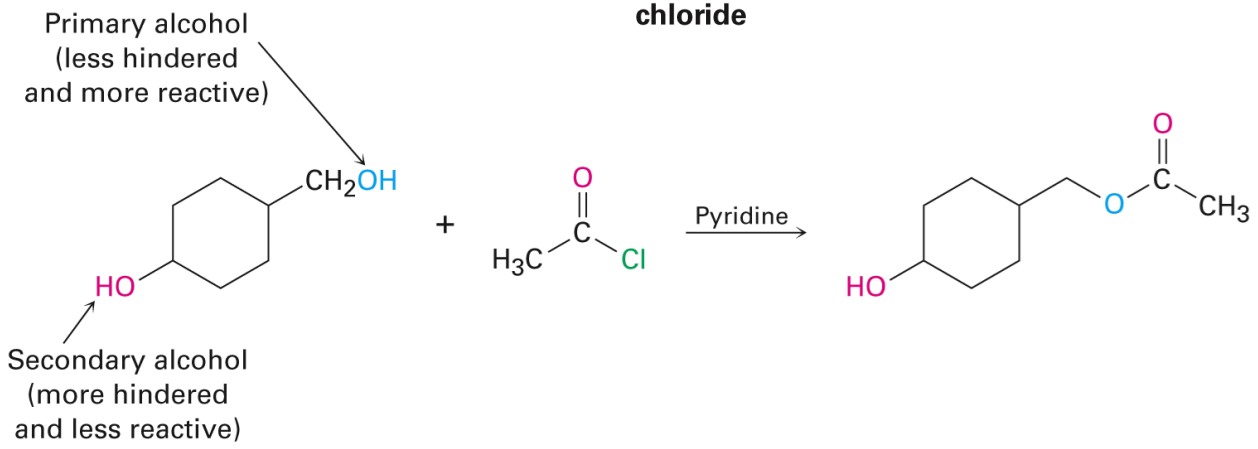
Sodium formate **Acetyl chloride** **Acetic formic anhydride (64%)**

Chemistry of Acid Halides

- Conversion of acid halides into esters: Alcoholysis
  - Esters are produced in the reaction of acid chlorides with alcohols in the presence of pyridine or NaOH

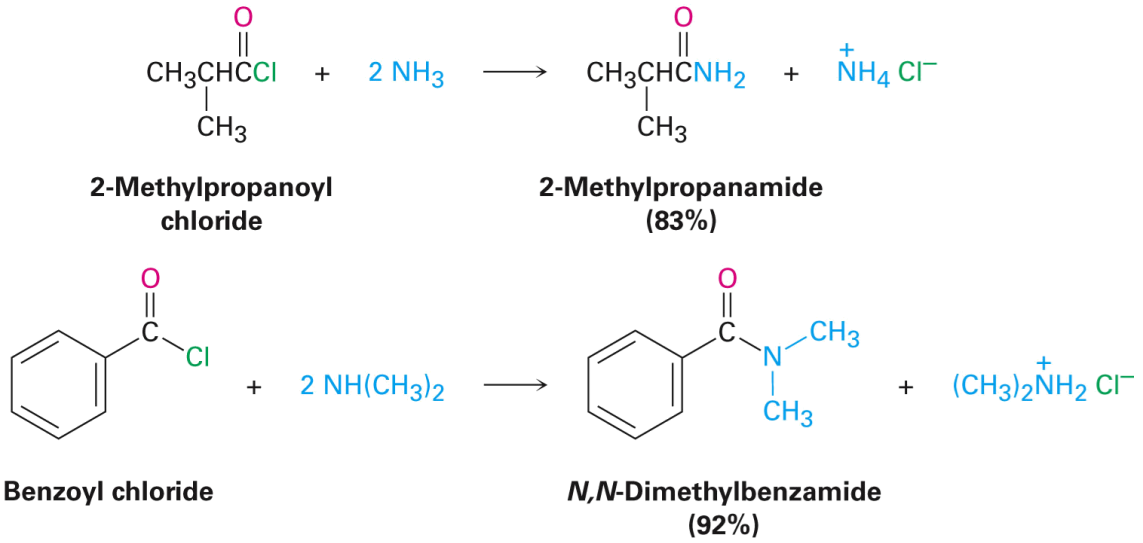


Benzoyl chloride **Cyclohexanol** **Cyclohexyl benzoate (97%)**



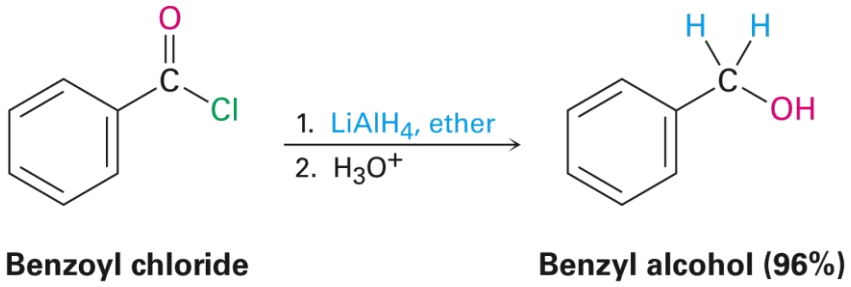
Chemistry of Acid Halides

- Aminolysis
  - Conversion of acid halides into amides acid chloride-plus-alcohol method is used to prepare esters
    - Method is used in preparation of amines
  - Trisubstituted amines(R_3N) cannot be used



Chemistry of Acid Halides

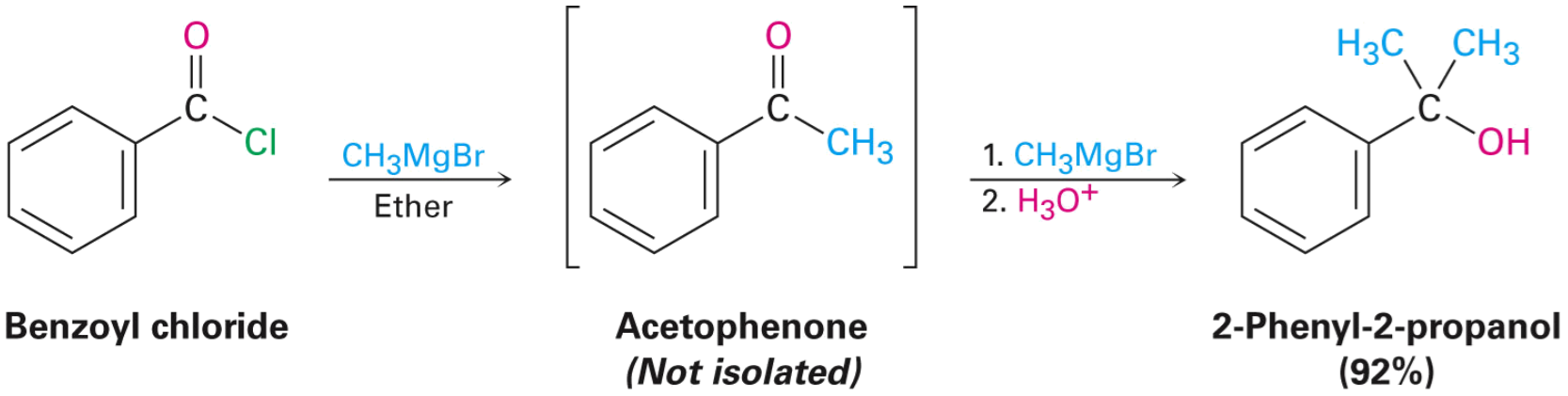
- Conversion of acid chlorides into alcohols: Reduction and Grignard reaction
  - $LiAlH_4$ reduces acid chlorides to yield aldehydes and then primary alcohols
  - Reduction occurs via nucleophilic acyl substitution mechanism



Benzoyl chloride **Benzyl alcohol (96%)**

Chemistry of Acid Halides

- Grignard reagents react with acid chlorides to yield tertiary alcohols with two identical substituent
- Reduction occurs via nucleophilic acyl substitution mechanism



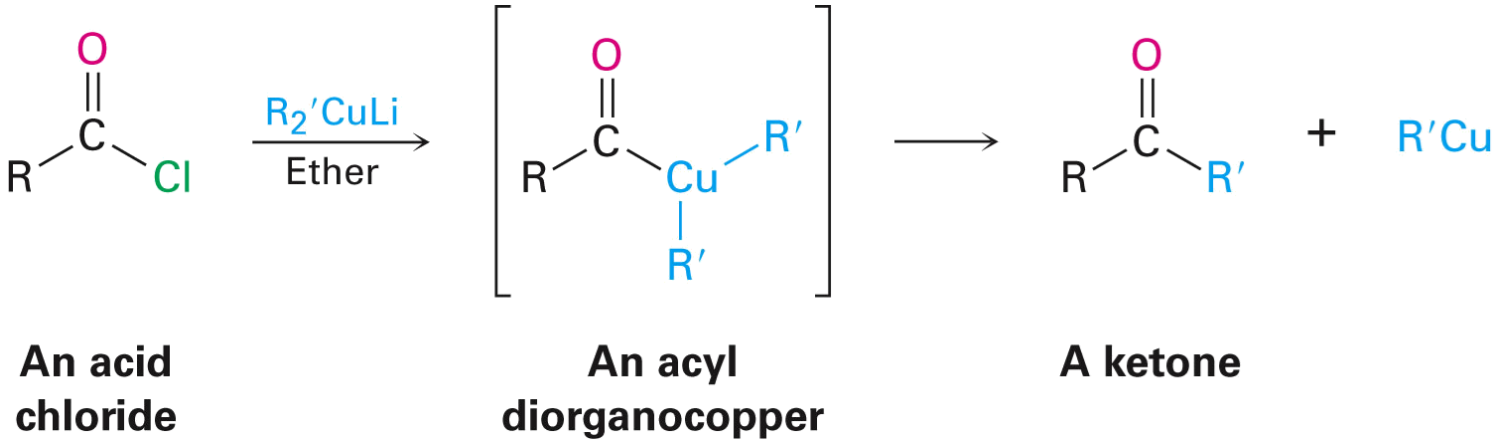
Benzoyl chloride

Acetophenone
(Not isolated)

2-Phenyl-2-propanol
(92%)

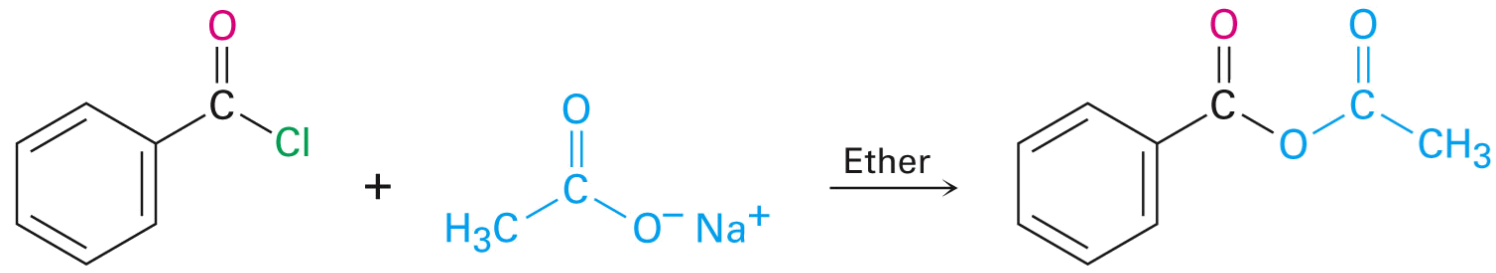
Formation of Ketones from Acid Chlorides

- Conversion of acid chlorides into ketones: Diorganocopper reaction
  - Reaction of an acid chloride with a lithium diorganocopper (Gilman) reagent, $Li^+ R'_2Cu^-$
  - Addition produces an acyl diorganocopper intermediate, followed by loss of R′Cu and formation of the ketone



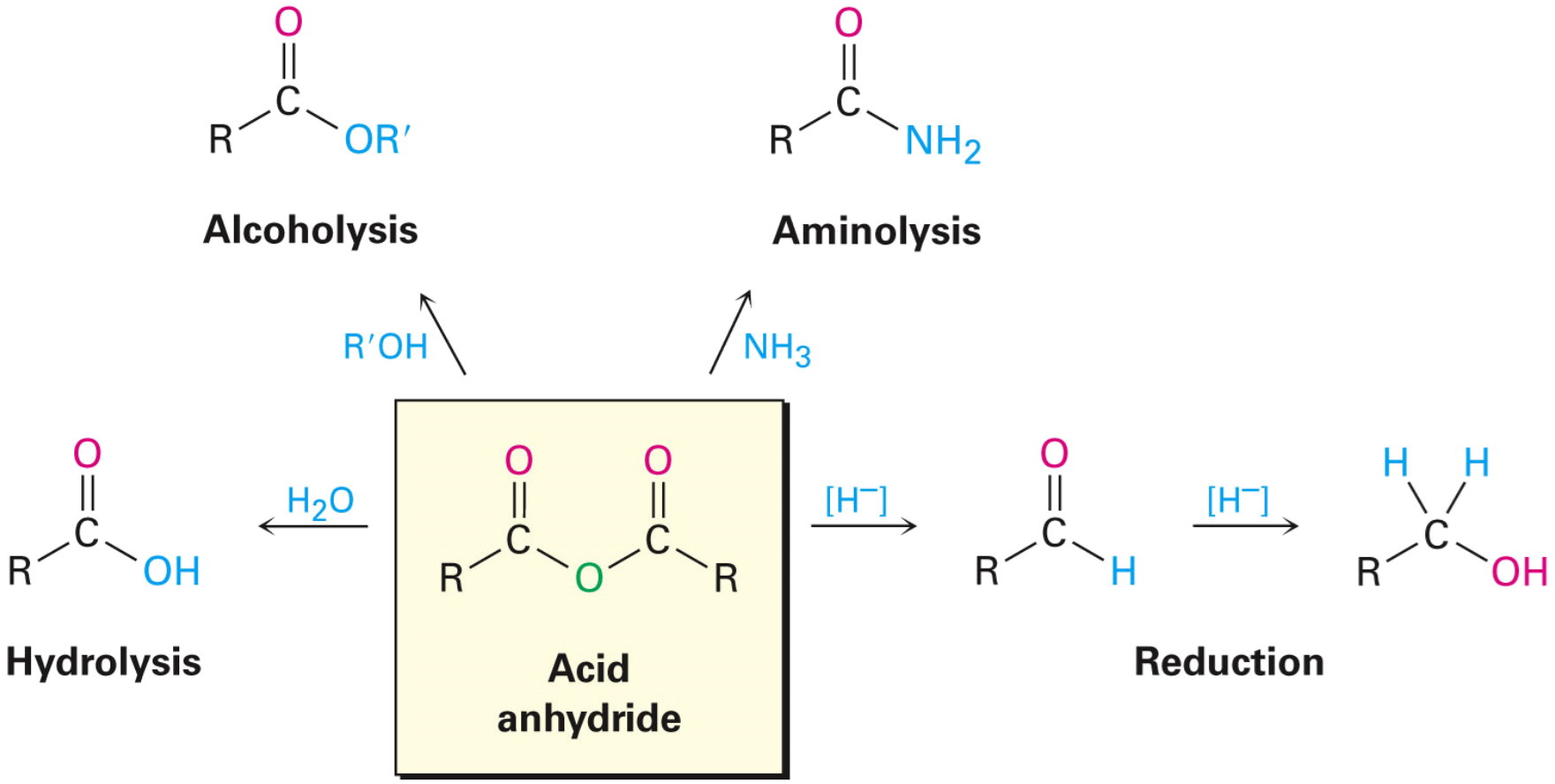
Chemistry of Acid Anhydrides

- Preparation of acid anhydrides
  - Nucleophilic acyl substitution of a carboxylate with an acid chloride



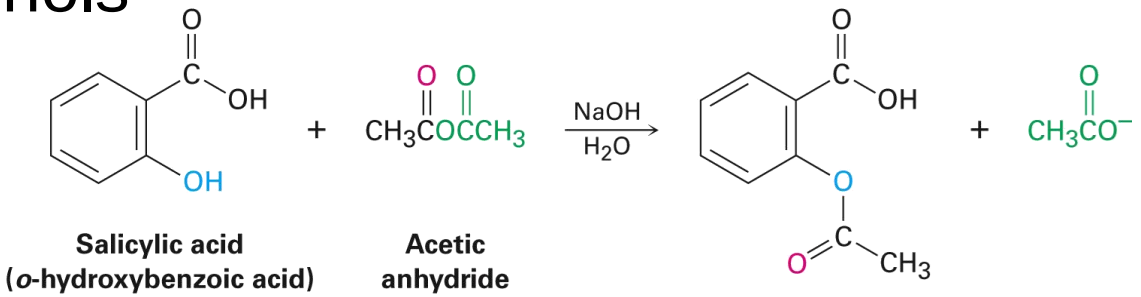
Benzoyl chloride **Sodium acetate** **Acetic benzoic anhydride**

Reactions of Acid Anhydrides



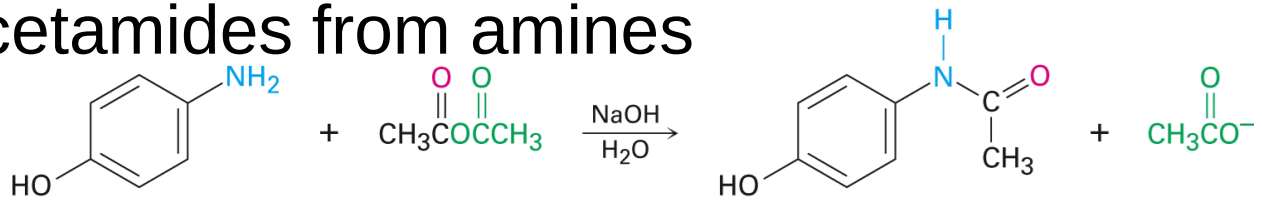
Reactions of Acid Anhydrides

- Conversion of acid anhydrides into esters
  - Acetic anhydride forms acetate esters from alcohols



Salicylic acid (*o*-hydroxybenzoic acid) | Acetic anhydride | Aspirin (an ester)

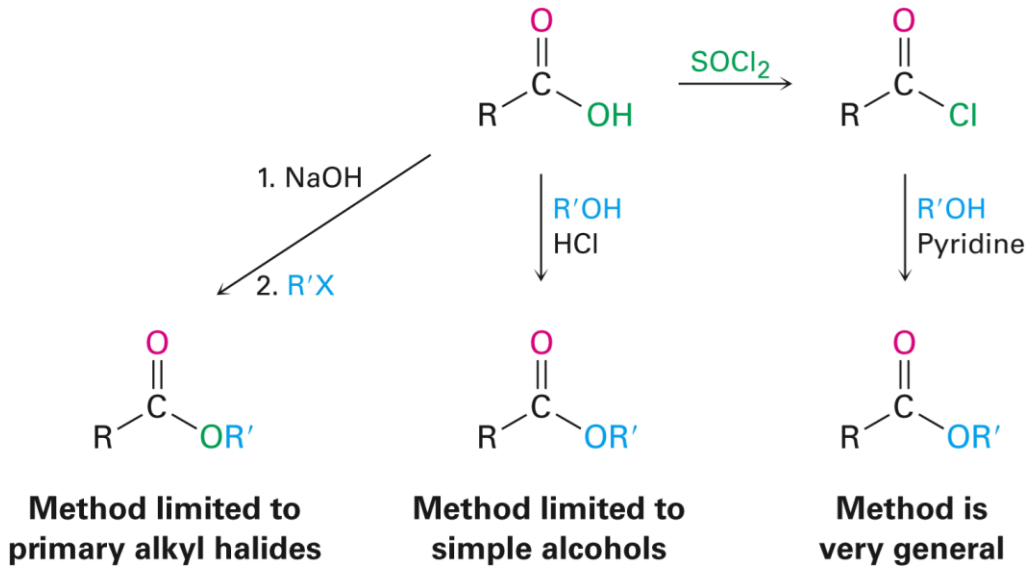
- Conversion of acid anhydrides into amides
  - Acetic anhydride is used to prepare N-substituted acetamides from amines



p-Hydroxyaniline | Acetic anhydride | Acetaminophen

Chemistry of Esters

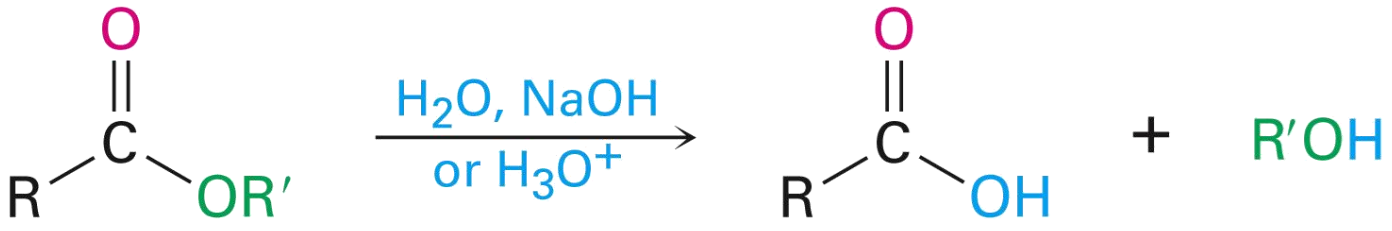
- Preparation of esters
  - Esters are usually prepared from carboxylic acids



  - Acid chlorides are converted into esters by treatment with an alcohol in the presence of base

Chemistry of Esters

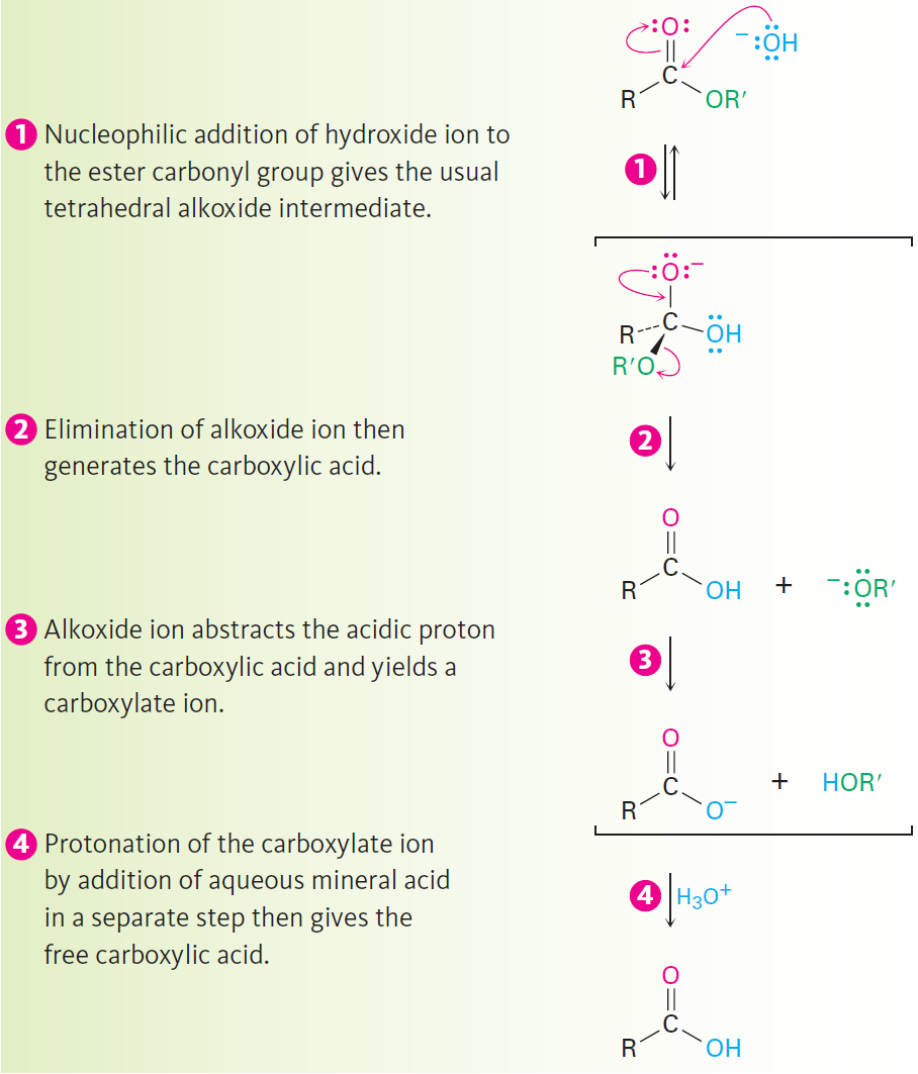
- Conversion of esters into carboxylic acids: Hydrolysis
  - An ester is hydrolyzed by aqueous base or aqueous acid to yield a carboxylic acid plus an alcohol



Ester **Acid** **Alcohol**

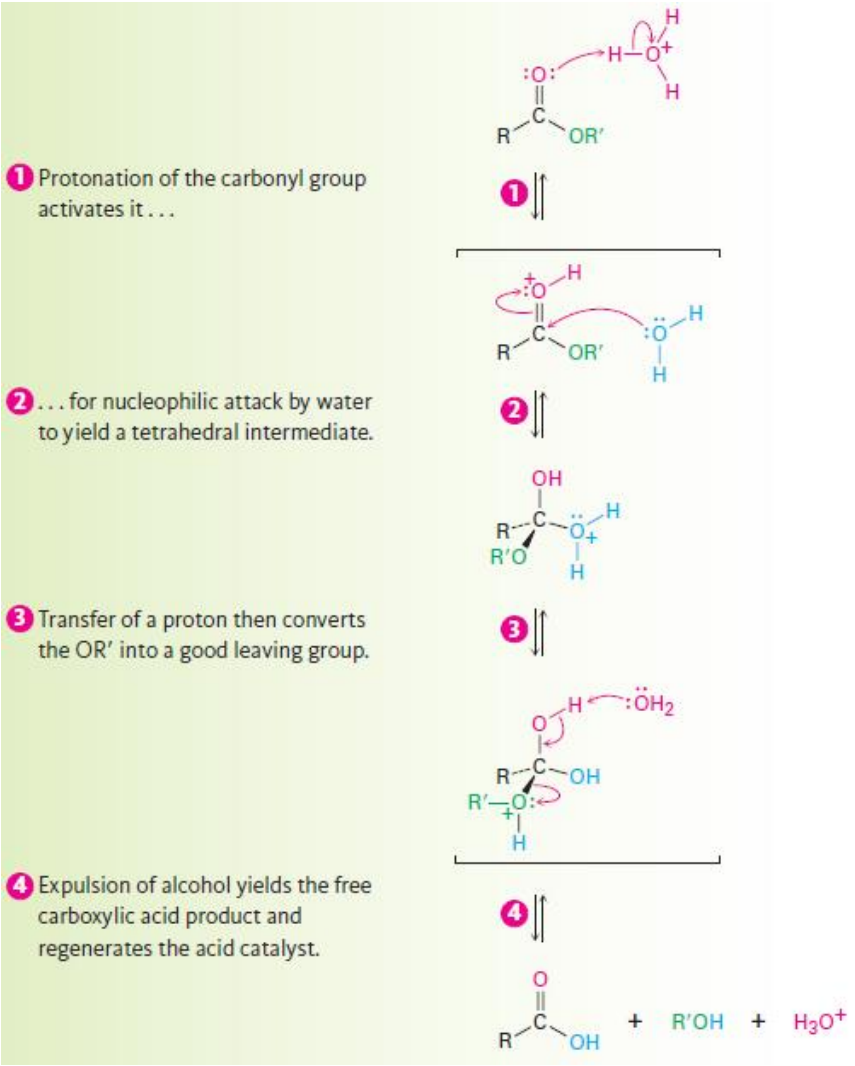
  - **Saponification:** Ester hydrolysis in basic solution

Figure 21.7 - Mechanism of Base-induced Ester Hydrolysis



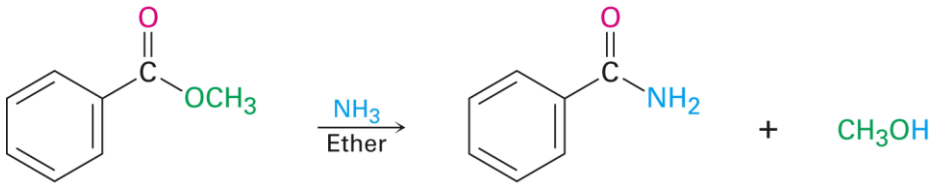
Chemistry of Esters

- Hydrolysis: Conversion of esters into carboxylic acids
  - Acid-catalyzed ester hydrolysis can occur by different mechanisms
    - Depends on the structure of the ester



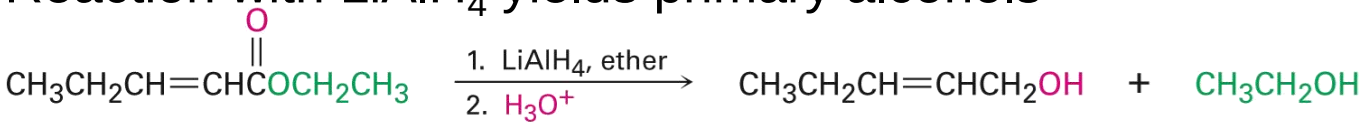
Chemistry of Esters

- Conversion of esters into amides: Aminolysis
  - Ammonia reacts with esters to form amides

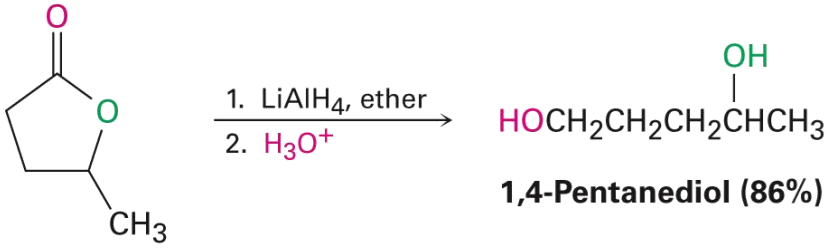


Methyl benzoate **Benzamide**

- Conversion of esters into alcohols: Reduction
  - Reaction with $LiAlH_4$ yields primary alcohols



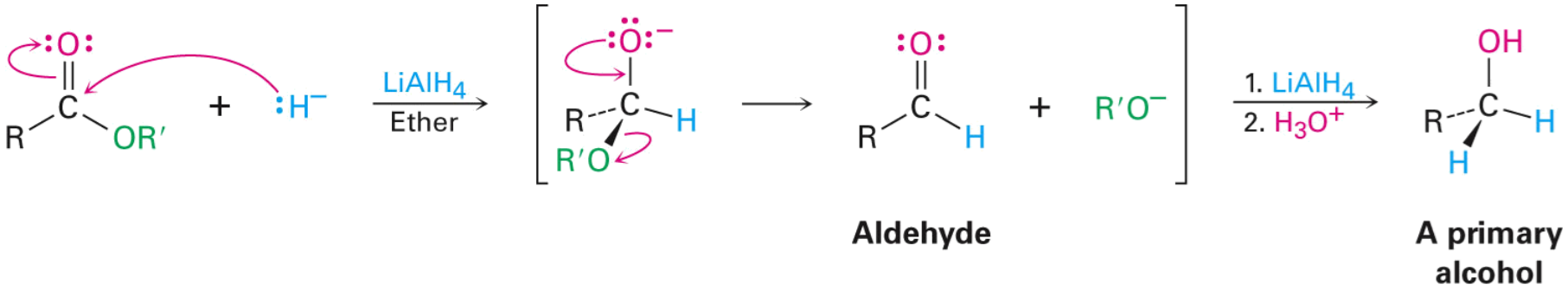
Ethyl 2-pentenoate **2-Penten-1-ol (91%)**



A lactone **1,4-Pentanediol (86%)**

Chemistry of Esters

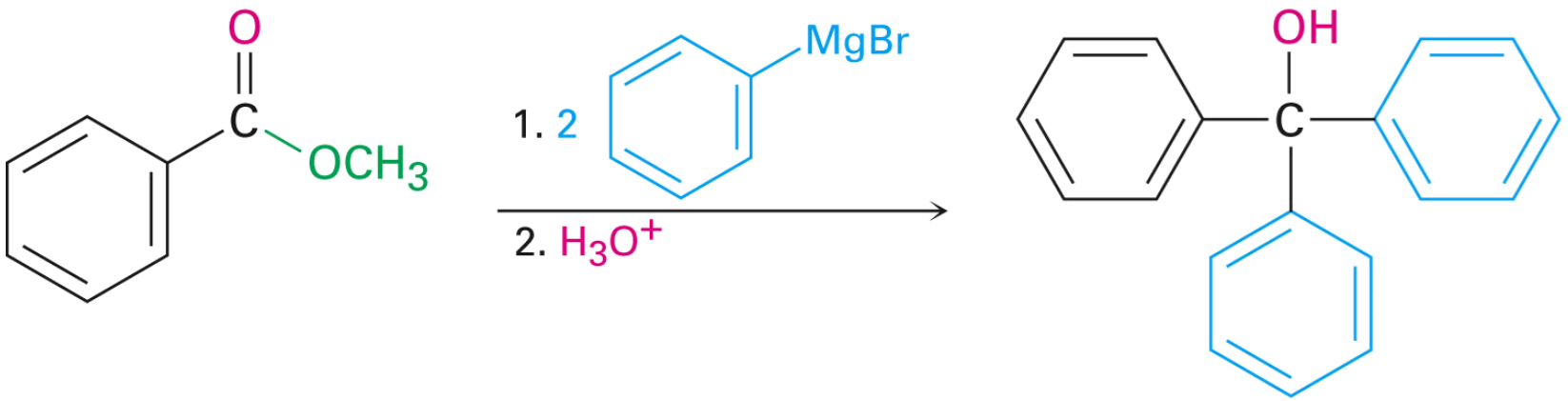
- Hydride ion adds to the carbonyl group, followed by elimination of alkoxide ion to yield an aldehyde
- Reduction of the aldehyde gives the primary alcohol



- Aldehyde intermediate can be isolated if 1 equivalent of diisobutylaluminum hydride (DIBAH, or DIBAL-H) is used as reducing agent

Chemistry of Esters

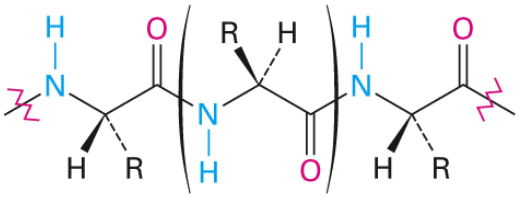
- Conversion of esters into alcohols: Grignard reaction
  - Esters react with two equivalents of a Grignard reagent to yield a tertiary alcohol



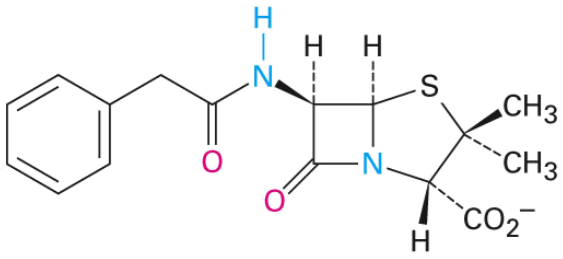
Methyl benzoate **Triphenylmethanol (96%)**

Chemistry of Amides

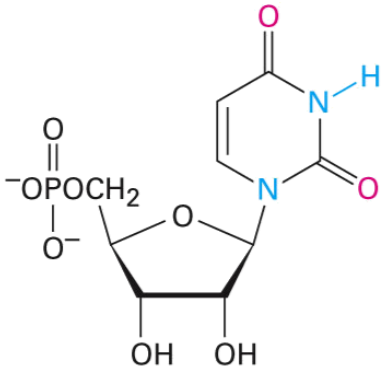
- Amides are abundant in living organisms
- Proteins, nucleic acids, and other pharmaceuticals have amide functional groups
- Amides are the least reactive of the common acid derivative



A protein segment



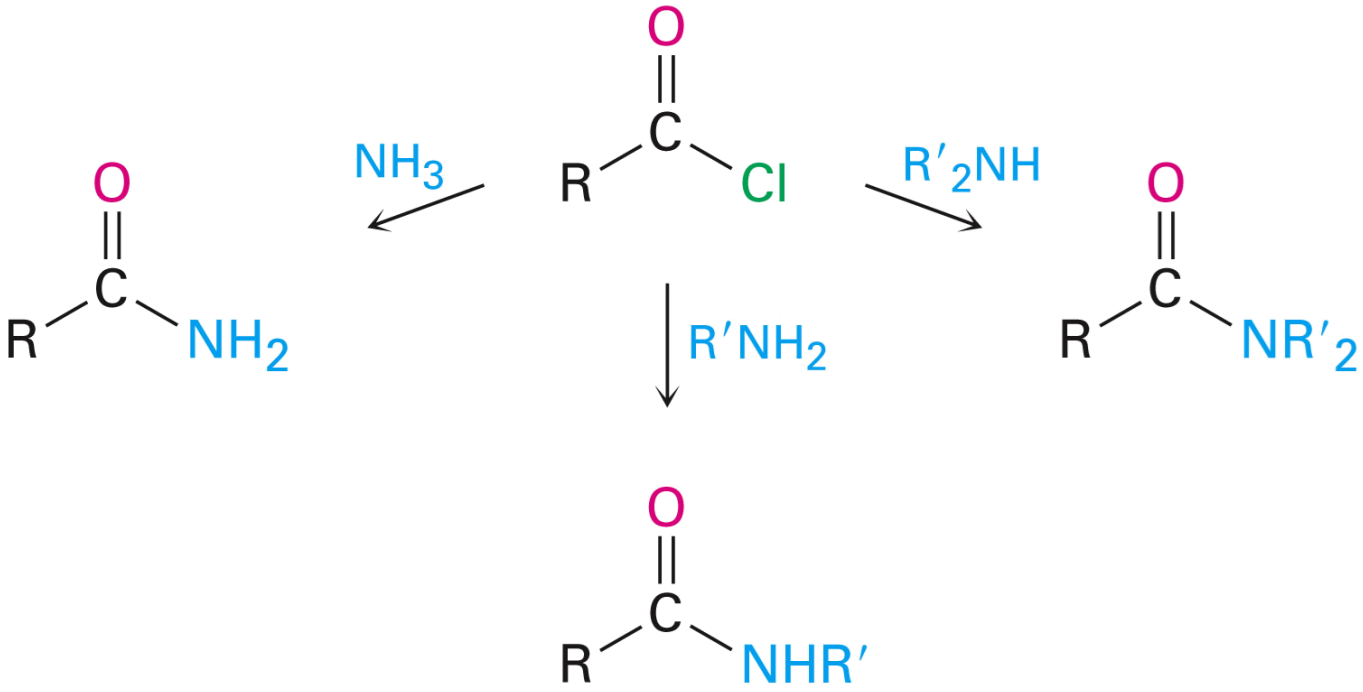
Benzylpenicillin
(penicillin G)



Uridine 5′-phosphate
(a ribonucleotide)

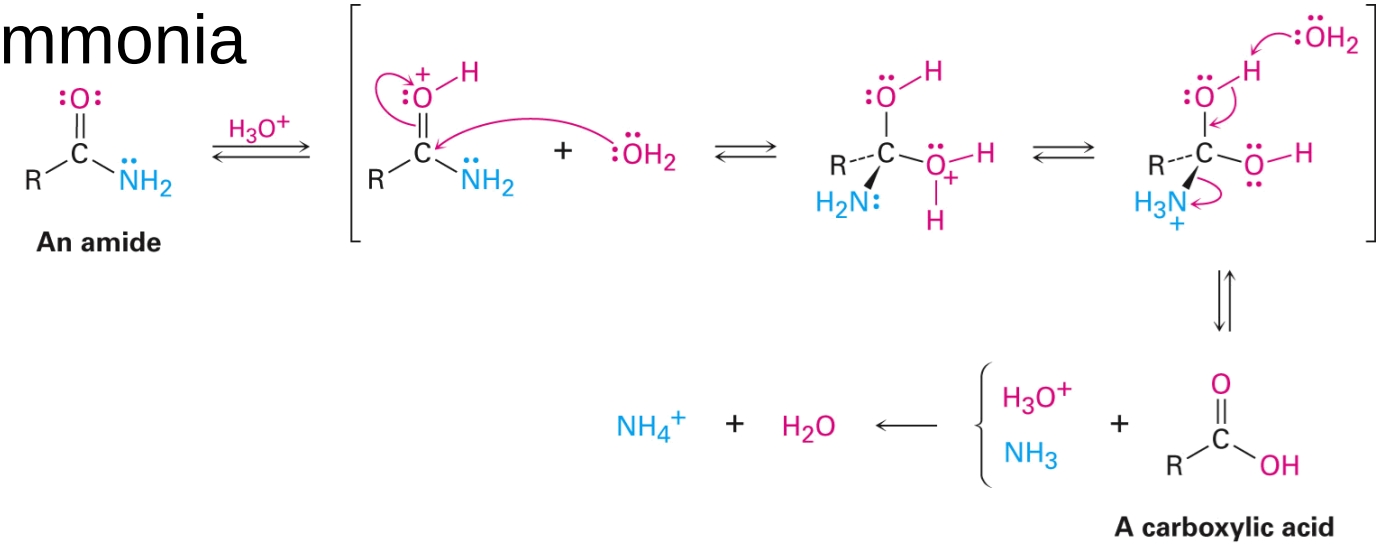
Preparation of Amides

- Chemistry of amides
  - Prepared by reaction of an acid chloride with ammonia, monosubstituted amines, or disubstituted amines



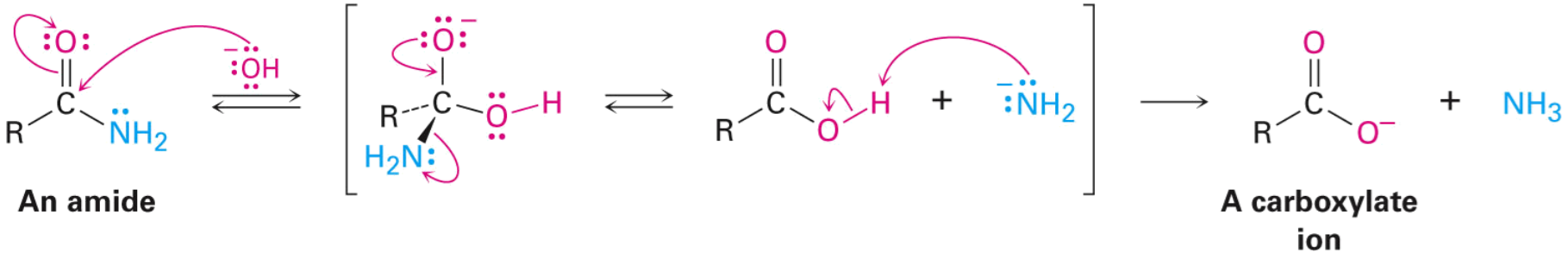
Reactions of Amides

- Conversion of amides into carboxylic acids: Hydrolysis
  - Heating in either aqueous acid or aqueous base produces a carboxylic acid and amine
  - Acidic hydrolysis by nucleophilic addition of water to the protonated amide, followed by loss of ammonia



Reactions of Amides

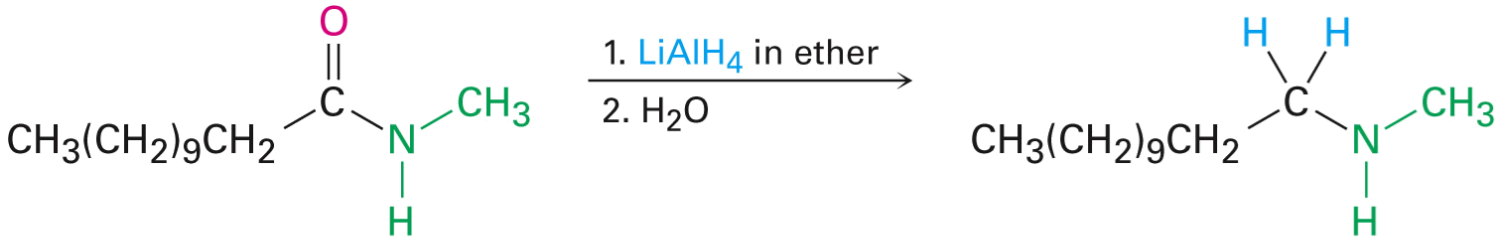
- Hydrolysis is difficult in comparison to analogous acid-catalyzed reaction because amide ion is a very poor leaving group
- Addition of hydroxide and loss of amide ion



- In biological chemistry, amide hydrolysis is common

Reactions of Amides

- Conversion of amides into amines: Reduction
  - Reduced by $LiAlH_4$ to an amine rather than an alcohol
  - Converts C=O $\rightarrow$ CH_2

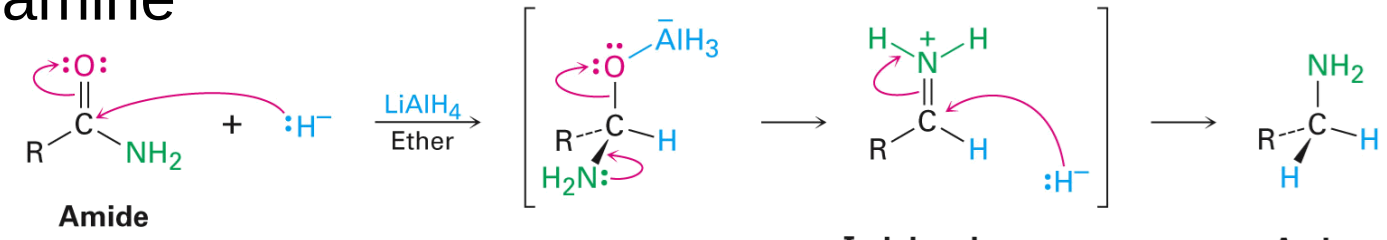


***N*-Methyldodecanamide**

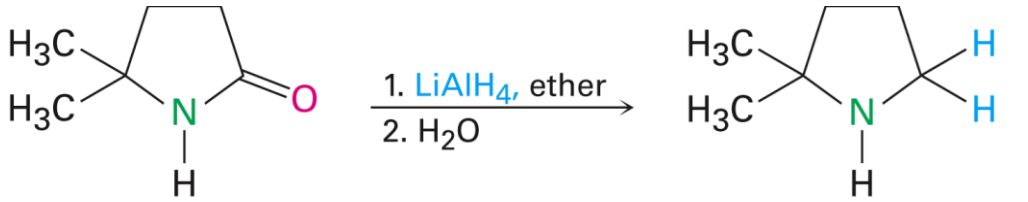
Dodecylmethylamine (95%)

Reactions of Amides

- Addition of hydride to carbonyl group
- Loss of the oxygen as an aluminate anion to give an iminium ion intermediate which is reduced to the amine

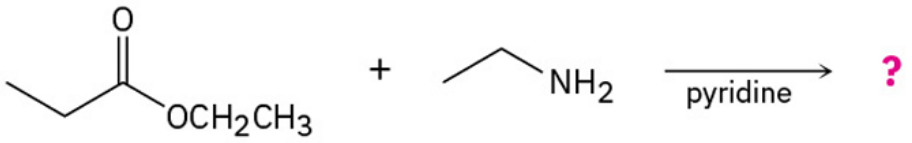


- The reaction is effective with both acyclic and cyclic amides, or **lactams**
- Good route for preparing cyclic amines

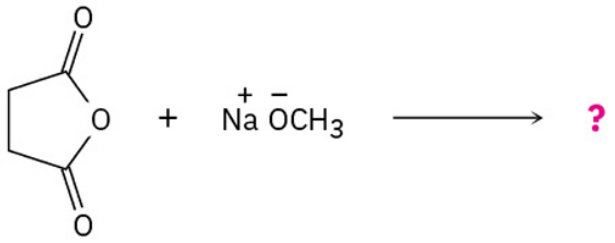


PROBLEM 21-31 Predict the product(s) and write the mechanism for the following reactions:

(a)

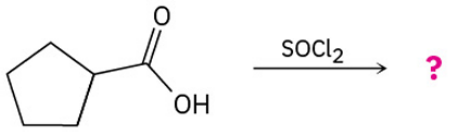


(b)

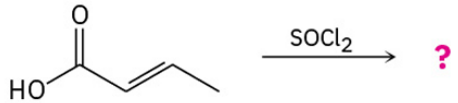


PROBLEM 21-32 Predict the product(s) and write the mechanism for the following reactions:

(a)

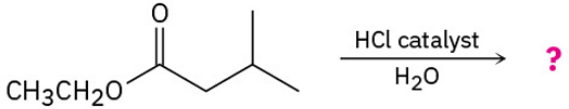


(b)

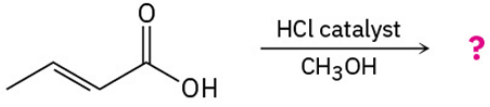


PROBLEM 21-33 Predict the product(s) and write the mechanism for each of the following reactions:

(a)

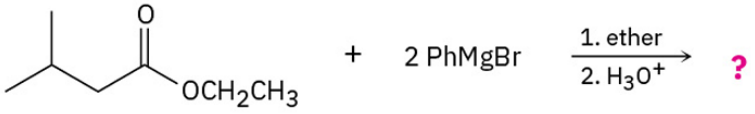


(b)

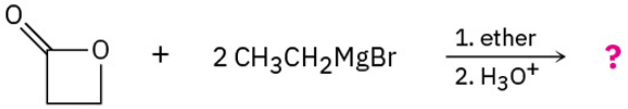


PROBLEM 21-34 Predict the product(s) and write the mechanism for the following reactions:

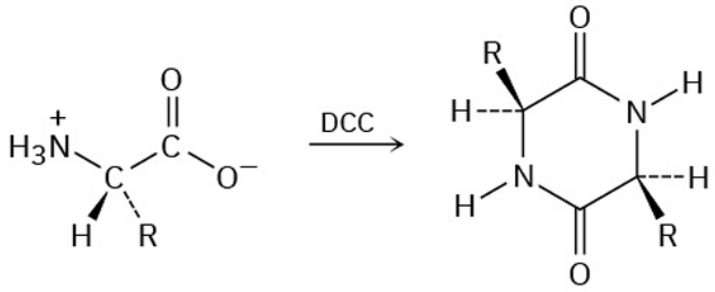
(a)



(b)



PROBLEM 21-38 Treatment of an α-amino acid with DCC yields a 2,5-diketopiperazine. Propose a mechanism.

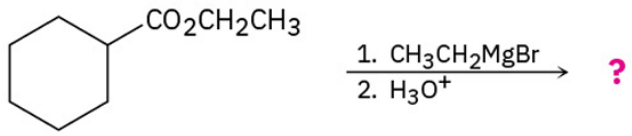


An α-amino acid **A 2,5-diketopiperazine**

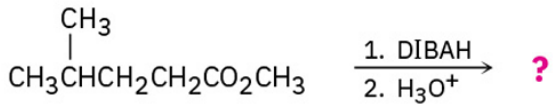
PROBLEM 21-48 Predict the product, if any, of reaction between propanoyl chloride and the following reagents: **(a)** $Li(Ph)_2Cu$ in ether **(b)** $LiAlH_4$, then H_3O^+ **(c)** CH_3MgBr, then H_3O^+ **(d)** H_3O^+ **(e)** Cyclohexanol **(f)** Aniline **(g)** $CH_3CO_2^-\ Na^+$

PROBLEM 21-53 Predict the product(s) of the following reactions:

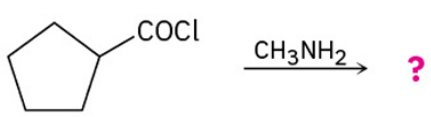
(a)



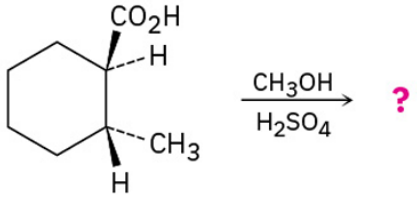
(b)



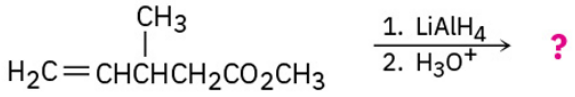
(c)



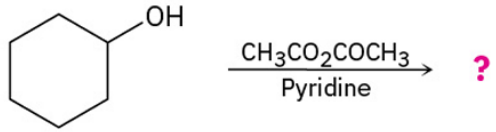
(d)



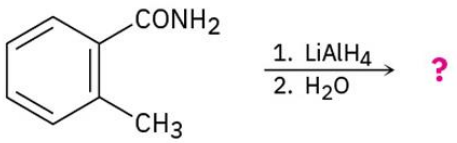
(e)



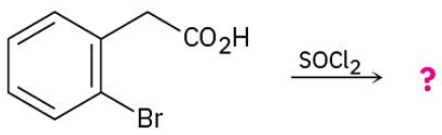
(f)



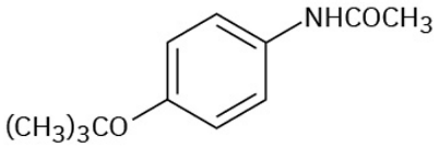
(g)



(h)

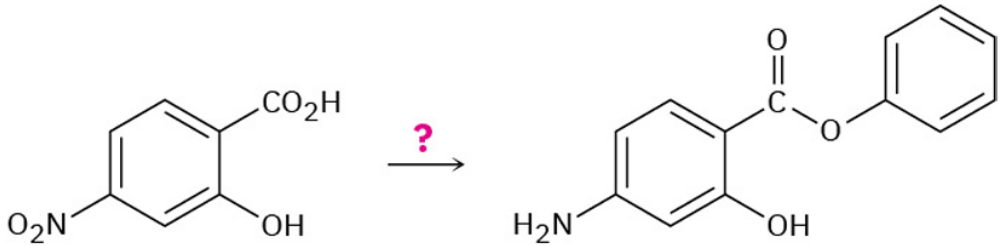


PROBLEM 21-71 Butacetin is an analgesic (pain-killing) agent that is synthesized commercially from *p*-fluoronitrobenzene. Propose a synthesis.



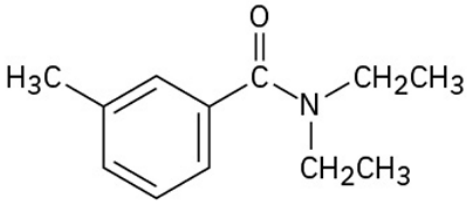
Butacetin

PROBLEM 21-72 Phenyl 4-aminosalicylate is a drug used in the treatment of tuberculosis. Propose a synthesis of this compound starting from 4-nitrosalicylic acid.



4-Nitrosalicylic acid **Phenyl 4-aminosalicylate**

PROBLEM 21-73 *N,N*-Diethyl-*m*-toluamide (DEET) is the active ingredient in many insect-repellent preparations. How might you synthesize this substance from *m*-bromotoluene?



***N,N*-Diethyl-*m*-toluamide**